

Java Study Material - Beginner to Advanced

PHASE 1: Basics of Java (Beginner Level)

1. Introduction to Java

- What is Java?
- Features of Java
- Java Development Kit (JDK), Java Runtime Environment (JRE), Java Virtual Machine (JVM)
- How Java works

2. Setting Up Java

- Installing JDK
- Setting up environment variables
- Installing an IDE (e.g., IntelliJ IDEA or Eclipse)

3. Hello World Program

- Writing your first program
- Compiling and running Java programs

4. Java Syntax and Structure

- Classes and Objects
- Main method
- Comments (Single-line and Multi-line)

5. Data Types and Variables

- Primitive data types
- Type casting
- Variables and constants

6. Operators

- Arithmetic, Relational, Logical, Bitwise, Assignment, Unary, Ternary

7. Control Flow Statements

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- if, if-else, nested if
- switch-case
- for, while, do-while loops
- break, continue

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PHASE 2: Object-Oriented Programming (OOP)

1. Classes and Objects
2. Constructors
3. this Keyword
4. Inheritance
5. Method Overloading & Overriding
6. Encapsulation
7. Abstraction
8. Polymorphism

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PHASE 3: Intermediate Java Concepts

1. Arrays and Strings
2. Static Keyword
3. Final Keyword
4. Access Modifiers
5. Packages and Import
6. Exception Handling
 - try-catch
 - throw, throws, finally
 - Custom exceptions

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PHASE 4: Advanced Java (Core Focus)

1. Wrapper Classes
2. Collections Framework
 - List, Set, Map
 - ArrayList, LinkedList, HashSet, TreeSet, HashMap, TreeMap
3. Generics
4. Lambda Expressions
5. Multithreading
 - Thread class and Runnable interface
 - Lifecycle of a thread
6. File Handling

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PHASE 5: GUI and Java Development

1. Introduction to GUI in Java
 - Swing basics
 - JFrame, JButton, JLabel, TextField, etc.
2. JavaFX (optional)
3. Mini Project: Build a simple calculator or student management system

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Next Step: Video Tutorials (After Text-Based Study)

Once you're comfortable with the concepts above, I'll recommend a list of YouTube playlists and online Java courses (free/paid) for visual learning.